Chapter 19 Oxidative Phosphorylation and Photophosphorylation

Multiple Choice Questions

1. Almost all of the oxygen (O_2) one consumes in breathing is converted to:

   A) acetyl-CoA.
   B) carbon dioxide (CO_2).
   C) carbon monoxide and then to carbon dioxide.
   D) None of the above
   E) water.

   Electron-transfer reactions in mitochondria
   Page: 732 Difficulty: 1 Ans: E

2. Which of the following electron carriers is *not* able to transfer one electron at a time?

   A) NADH
   B) FMN
   C) FAD
   D) Ubiquinone
   E) Heme

   Electron-transfer reactions in mitochondria
   Pages: 734–736 Difficulty: 1 Ans: A

3. A new compound isolated from mitochondria is claimed to represent a previously unrecognized carrier in the electron transfer chain. It is given the name coenzyme Z. Which line of evidence do you feel is the *least* conclusive in assigning this compound a position in the electron transfer chain?

   A) Alternate oxidation and reduction of the mitochondrion-bound coenzyme Z can be readily demonstrated.
   B) Removal of coenzyme Z from the mitochondria results in a decreased rate of oxygen consumption.
   C) The rate of oxidation and reduction of mitochondrion-bound coenzyme is of the same order of magnitude as the overall rate of electron transfer in mitochondria as measured by oxygen consumption.
   D) The reduction potential of Z is between that of two compounds known to participate in the electron transport chain
   E) When added to a mitochondrial suspension, coenzyme Z is taken up very rapidly and specifically by the mitochondria.

   Electron-transfer reactions in mitochondria
   Pages: 735–737Difficulty: 3 Ans: E

4. Antimycin A blocks electron transfer between cytochromes *b* and c_1. If intact mitochondria were

incubated with antimycin A, excess NADH, and an adequate supply of O_2, which of the following would be found in the oxidized state?

A) Coenzyme Q
B) Cytochrome a_3
C) Cytochrome *b*
D) Cytochrome *e*
E) Cytochrome *f*

Electron-transfer reactions in mitochondria
Pages: 735–737Difficulty: 2 Ans: B

5. Reduced QH_2 is *not* formed by which of the following?

A) Complex I and NADH
B) Complex II and succinate
C) Complex III and cytochrome *c*
D) Fatty acid oxidation
E) Oxidation of glycerol-3-phosphate

Electron-transfer reactions in mitochondria
Page: 739 Difficulty: 2 Ans: C

6. In the reoxidation of QH_2 by purified ubiquinone-cytochrome *c* reductase (Complex III) from heart muscle, the overall stoichiometry of the reaction requires 2 mol of cytochrome *c* per mole of QH_2 because:

A) cytochrome *c* is a one-electron acceptor, whereas QH_2 is a two-electron donor.
B) cytochrome *c* is a two-electron acceptor, whereas QH_2 is a one-electron donor.
C) cytochrome *c* is water soluble and operates between the inner and outer mitochondrial membranes
D) heart muscle has a high rate of oxidative metabolism, and therefore requires twice as much cytochrome *c* as QH_2 for electron transfer to proceed normally.
E) two molecules of cytochrome *c* must first combine physically before they are catalytically active.

Electron-transfer reactions in mitochondria
Pages: 740–742 Difficulty: 2 Ans: A

7. Which of the following is *not* a feature of complex IV?

A) Cytochrome *c* is a one-electron donor.
B) Oxygen is a substrate.
C) Copper is an essential metal for the reaction.
D) For every electron passed to complex IV, two protons are consumed from the matrix (N) side.
E) In order to generate two water molecules, complex IV must go through the catalytic cycle two times.

Electron-transfer reactions in mitochondria

Pages: 741–743 Difficulty: 2 Ans: E

8. Which of the following is *not* true of the proton motive force (pmf)?

   A) One component of the pmf is the chemical gradient of protons.
   B) One component of the pmf is the charge gradient of protons.
   C) Generation of the pmf in mitochondria requires succinate.
   D) The pmf is generated by the electron transport chain in mitochondria.
   E) The pmf drives ATP synthesis in mitochondria.

Electron-transfer reactions in mitochondria
Pages: 744–745 Difficulty: 2 Ans: C

9. Cyanide, oligomycin, and 2,4-dinitrophenol (DNP) are inhibitors of mitochondrial aerobic phosphorylation. Which of the following statements correctly describes the mode of action of the three inhibitors?

   A) Cyanide and 2,4-dinitrophenol inhibit the respiratory chain, and oligomycin inhibits the synthesis of ATP.
   B) Cyanide inhibits the respiratory chain, whereas oligomycin and 2,4-dinitrophenol inhibit the synthesis of ATP.
   C) Cyanide, oligomycin, and 2,4-dinitrophenol compete with O_2 for cytochrome oxidase (complex IV).
   D) Oligomycin and cyanide inhibit synthesis of ATP; 2,4-dinitrophenol inhibits the respiratory chain.
   E) Oligomycin inhibits the respiratory chain, whereas cyanide and 2,4-dinitrophenol prevent the synthesis of ATP.

ATP synthesis
Pages: 739, 748–749 Difficulty: 2 Ans: B

10. If electron transfer in tightly coupled mitochondria is blocked (with antimycin A) between cytochrome *b* and cytochrome c_1, then:

   A) all ATP synthesis will stop.
   B) ATP synthesis will continue, but the P/O ratio will drop to one.
   C) electron transfer from NADH will cease, but O_2 uptake will continue.
   D) electron transfer from succinate to O_2 will continue unabated.
   E) energy diverted from the cytochromes will be used to make ATP, and the P/O ratio will rise.

ATP synthesis
Pages: 748–749 Difficulty: 3 Ans: A

11. In normal mitochondria, the rate of NADH consumption (oxidation) will:

   A) be increased in active muscle, decreased in inactive muscle.
   B) be very low if the ATP synthase is inhibited, but increase when an uncoupler is added.
   C) decrease if mitochondrial ADP is depleted.
   D) decrease when cyanide is used to prevent electron transfer through the cytochrome $a + a_3$ complex.
   E) All of the above

ATP synthesis
Pages: 748–749 Difficulty: 3 Ans: E

12. Which of the following statements about the chemiosmotic theory is *correct*?

A) Electron transfer in mitochondria is accompanied by an asymmetric release of protons on *one* side of the inner mitochondrial membrane.
B) It predicts that oxidative phosphorylation can occur, even in the absence of an intact inner mitochondrial membrance.
C) The effect of uncoupling reagents is a consequence of their ability to carry electrons through membranes.
D) The membrane ATP synthase has no significant role in the chemiosmotic theory.
E) All of the above

ATP synthesis
Pages: 748–749 Difficulty: 2 Ans: A

13. Which of the following statements about the chemiosmotic theory is *false*?

A) Electron transfer in mitochondria is accompanied by an asymmetric release of protons on *one* side of the inner mitochondrial membrane.
B) Energy is conserved as a transmembrane pH gradient.
C) Oxidative phosphorylation cannot occur in membrane-free preparations.
D) The effect of uncoupling reagents is a consequence of their ability to carry protons through membranes.
E) The membrane ATPase, which plays an important role in other hypotheses for energy coupling, has no significant role in the chemiosmotic theory.

ATP synthesis
Pages: 748–749 Difficulty: 1 Ans: E

14. Upon the addition of 2,4-dinitrophenol (DNP) to a suspension of mitochondria carrying out oxidative phosphorylation linked to the oxidation of malate, all of the following occur *except*:

A) oxygen consumption decreases.
B) oxygen consumption increases.
C) the P/O ratio drops from a value of approximately 2.5 to 0.
D) the proton gradient dissipates.
E) the rate of transport of electrons from NADH to O_2 becomes maximal.

ATP synthesis
Pages: 748–749 Difficulty: 3 Ans: A

15. Uncoupling of mitochondrial oxidative phosphorylation:

A) allows continued mitochondrial ATP formation but halts O_2 consumption.
B) halts all mitochondrial metabolism.
C) halts mitochondrial ATP formation but allows continued O_2 consumption.
D) slows down the citric acid cycle.
E) slows the conversion of glucose to pyruvate by glycolysis.

ATP synthesis
Pages: 748–749 Difficulty: 1 Ans: C

16. 2,4-Dinitrophenol and oligomycin inhibit mitochondrial oxidative phosphorylation. 2,4-Dinitrophenol is an uncoupling agent; oligomycin blocks the ATP synthesis reaction itself. Therefore, 2,4-dinitrophenol will:

A) allow electron transfer in the presence of oligomycin.
B) allow oxidative phosphorylation in the presence of oligomycin.
C) block electron transfer in the presence of oligomycin.
D) diminish O_2 consumption in the presence of oligomycin.
E) None of the above

ATP synthesis
Pages: 748–749 Difficulty: 2 Ans: A

17. Which of the following statements about energy conservation in the mitochondrion is *false*?

A) Drugs that inhibits the ATP synthase will also inhibit the flow of electrons down the chain of carriers.
B) For oxidative phosphorylation to occur, it is essential to have a closed membranous structure with an inside and an outside.
C) The yield of ATP per mole of oxidizable substrate depends on the substrate.
D) Uncouplers (such as dinitrophenol) have exactly the same effect on electron transfer as inhibitors such as cyanide; both block further electron transfer to oxygen.
E) Uncouplers "short circuit" the proton gradient, thereby dissipating the proton motive force as heat.

ATP synthesis
Pages: 748–749 Difficulty: 1 Ans: D

18. Which of the following is *correct* concerning the mitochondrial ATP synthase?

A) It can synthesize ATP after it is extracted from broken mitochondria.
B) It catalyzes the formation of ATP even though the reaction has a large positive $\Delta G'^{\circ}$.
C) It consists of F_o and F_1 subunits, which are transmembrane (integral) polypeptides.
D) It is actually an ATPase and only catalyzes the hydrolysis of ATP.
E) When it catalyzes the ATP synthesis reaction, the $\Delta G'^{\circ}$ is actually close to zero.

ATP synthesis
Pages: 750–751 Difficulty: 2 Ans: E

19. When the $\Delta G'^{\circ}$ of the ATP synthesis reaction is measured on the surface of the ATP synthase enzyme, it is found to be close to zero. This is thought to be due to:

A) a very low energy of activation.
B) enzyme-induced oxygen exchange.
C) stabilization of ADP relative to ATP by enzyme binding.
D) stabilization of ATP relative to ADP by enzyme binding.
E) None of the above

ATP synthesis
Pages: 750–751 Difficulty: 2 Ans: D

20. During oxidative phosphorylation, the proton motive force that is generated by electron transport is used to:

A) create a pore in the inner mitochondrial membrane.
B) generate the substrates (ADP and P_i) for the ATP synthase.
C) induce a conformational change in the ATP synthase.
D) oxidize NADH to NAD^+.
E) reduce O_2 to H_2O.

ATP synthesis
Pages: 752–755 Difficulty: 2 Ans: C

21. The oxidation of a particular hydroxy substrate to a keto product by mitochondria has a P/O ratio of less than 2. The initial oxidation step is very likely directly coupled to the:

A) oxidation of a flavoprotein.
B) oxidation of a pyridine nucleotide.
C) reduction of a flavoprotein.
D) reduction of a pyridine nucleotide.
E) reduction of cytochrome a_3.

ATP synthesis
Pages: 755–757 Difficulty: 2 Ans: C

22. The relative concentrations of ATP and ADP control the cellular rates of:

A) glycolysis.
B) oxidative phosphorylation.
C) pyruvate oxidation.
D) the citric acid cycle.
E) All of the above

Regulation of oxidative phosphorylation
Page: 760 Difficulty: 2 Ans: E

23. The rate of oxidative phosphorylation in mitochondria is controlled primarily by:

A) feedback inhibition by CO_2.
B) the availability of NADH from the TCA cycle.
C) the concentration of citrate (or) the glycerol-3-phosphate shuttle.
D) the mass-action ratio of the ATD-ADP system.
E) the presence of thermogenin.

Regulation of oxidative phosphorylation

Page: 760 Difficulty: 2 Ans: D

24. Which of the following is *not* controlled by hypoxia-inducible factor (HIF-1)?

A) Glucose transport
B) Glycolysis
C) Citric acid cycle
D) Compex I of the respiratory chain
E) Complex IV of the respiratory chain

Regulation of oxidative phosphorylation
Page: 761 Difficulty: 2 Ans: D

25. Mammals produce heat by using the endogenous uncoupling agent:

A) the small molecule 2-4-Dinitrophenol synthesized by the cell.
B) the protein thermogenin.
C) the protein thioredoxin.
D) the protein cytochrome *c*.
E) a modified form of the F_oF_1 ATPase.

Mitochondria in thermogenesis, steroid synthesis, and apoptosis
Page: 763 Difficulty: 2 Ans: B

26. Which one of the following *best* describes the role of mitochondria in apoptosis?

A) Escape of cytochrome *c* into the cytoplasm
B) Increased rate of fatty acid β-oxidation
C) Increase in permeability of outer membrane
D) Uncoupling of oxidative phosphorylation
E) Both A and C are correct.

Mitochondria in thermogenesis, steroid synthesis, and apoptosis
Page: 764 Difficulty: 2 Ans: E

27. Mutations in mitochondrial genes play a role in each of the following diseases *except*:

A) adult onset diabetes.
B) cystic fibrosis.
C) hypertrophic cardiomyopathy.
D) Leber's hereditary optic neuropathy.
E) myoclonic epilepsy.

Mitochondrial genes: their origin and the effects of mutation
Pages: 767–768 Difficulty: 2 Ans: B

28. Which one of the following statements about human mitochondria is *true*?

A) About 900 mitochondrial proteins are encoded by nuclear genes.

B) Mitochondrial genes are inherited from both maternal and paternal sources.
C) rRNA and tRNA are imported from the cytoplasm and used in mitochondrial protein synthesis.
D) The mitochondrial genome codes for all proteins found in mitochondria.
E) The mitochondrial genome is not subject to mutations.

Mitochondrial genes: their origin and the effects of mutation
Pages: 765–768 Difficulty: 2 Ans: A

29. In photophosphorylation, absorption of light energy in chloroplast "light reactions" leads to:

A) absorption of CO_2 and release of O_2.
B) absorption of O_2 and release of CO_2.
C) hydrolysis of ATP and reduction of $NADP^+$.
D) synthesis of ATP and oxidation of NADPH.
E) use of iron-sulfur proteins.

General features of photophosphorylation
Page: 769 Difficulty: 2 Ans: A

30. Which of the following statements about the light reactions in photosynthetic plants is *false*?

A) A membrane-bound ATPase couples ATP synthesis to electron transfer.
B) No CO_2 is fixed in the light reactions.
C) The ultimate electron acceptor is O_2.
D) The ultimate source of electrons for the process is H_2O.
E) There are two distinct photosystems, linked together by an electron transfer chain.

General features of photophosphorylation
Pages: 769–770 Difficulty: 2 Ans: B

31. The light reactions in photosynthetic higher plants:

A) do not require chlorophyll.
B) produce ATP and consume NADH.
C) require the action of a single reaction center.
D) result in the splitting of H_2O, yielding O_2.
E) serve to produce light so that plants can see underground.

General features of photophosphorylation
Pages: 769–770 Difficulty: 1 Ans: D

32. Photosynthetic phosphorylation and oxidative phosphorylation appear to be generally similar processes, both consisting of ATP synthesis coupled to the transfer of electrons along an electron carrier chain. Which of the following is *not* true of both processes?

A) Both contain cytochromes and flavins in their electron carrier chains.
B) Both processes are associated with membranous elements of the cell.
C) Both use oxygen as a terminal electron acceptor.
D) Each represents the major route of ATP synthesis in those cells in which it is found.
E) Protons are pumped from the inside to the outside of both mitochondria and chloroplast membranes

General features of photophosphorylation
Pages: 769–770 Difficulty: 2 Ans: C

33. Oxidative phosphorylation and photophosphorylation share all of the following *except*:

A) chlorophyll.
B) involvement of cytochromes.
C) participation of quinones.
D) proton pumping across a membrane to create electrochemical potential.
E) use of iron-sulfur proteins.

General features of photophosphorylation
Pages: 769–770 Difficulty: 2 Ans: A

34. The experimental determination of the effectiveness of light of different colors in promoting photosynthesis is called the:

A) absorption spectrum.
B) action spectrum.
C) difference spectrum.
D) reflectance spectrum.
E) refraction spectrum.

Light absorption
Pages: 773–774 Difficulty: 2 Ans: B

35. In what order do the following five steps occur in the photochemical reaction centers?

1) Excitation of the chlorophyll *a* molecule at the reaction center
2) Replacement of the electron in the reaction center chlorophyll
3) Light excitation of antenna chlorophyll molecule
4) Passage of excited electron to electron-transfer chain
5) Exiton transfer to neighboring chlorophyll

A) 1-2-3-4-5
B) 3-2-5-4-1
C) 3-5-1-4-2
D) 4-2-3-5-1
E) 5-4-3-2-1

Light absorption
Pages: 774–775 Difficulty: 2 Ans: C

36. Which one of the following is *true* about reaction centers?

A) Cyanobacteria and plants have two reaction centers arranged in tandem.
B) Cyanobacteria contain a single reaction center of the Fe-S type.
C) Green sulfur bacteria have two reaction centers arranged in tandem.

D) Plant photosystems have a single reaction center of the pheophytin-quinone type.
E) Purple bacteria contain a single reaction center of the Fe-S type.

The central photochemical event: light-driven electron flow
Pages: 776–779 Difficulty: 2 Ans: A

37. In the photolytic cleavage of water by the oxygen-evolving complex [$2H_2O \rightarrow 4\ H^+ + 4e^- + O_2$], how many photons of light at a wavelength of 680 nm are required?

A) 1
B) 2
C) 4
D) 6
E) 8

The central photochemical event: light-driven electron flow
Page: 780 Difficulty: 2 Ans: C

38. Place the following electron-carriers into the proper order as found in plant chloroplasts.

1) Cytochrome b_6f complex
2) P680
3) P700
4) Plastocyanin
5) NADPH

A) 2, 1, 3, 4, 5
B) 2, 4, 3, 1, 5
C) 3, 1, 4, 2, 5
D) 2, 1, 4, 3, 5
E) 3, 4, 2, 1, 5

The central photochemical event: light-driven electron flow
Page: 779 Difficulty: 3 Ans: D

39. Which of the following is *not* true about cyclic electron flow?

A) Cyclic electron flow produces more NADPH per photon than noncyclic electron flow.
B) Cyclic electron flow involves only PSI.
C) Plastocyanin is required for cyclic electron flow.
D) Cyclic electron flow leads to the build-up of a proton gradient.
E) Cyclic electron flow does not produce O_2.

The central photochemical event: light-driven electron flow
Page: 783 Difficulty: 2 Ans: A

40. Cyclic electron flow in chloroplasts produces:

A) ATP and O_2, but not NADPH.
B) ATP, but not NADPH or O_2.

C) NADPH and ATP, but not O_2.
D) NADPH, but not ATP or O_2.
E) O_2, but not ATP or NADPH.

The central photochemical event: light-driven electron flow
Page: 783 Difficulty: 3 Ans: B

41. Which one of the following statements about photophosphorylation is *false*?

A) It can be uncoupled from electron flow by agents that dissipate the proton gradient.
B) The difference in pH between the luminal and stromal side of the thylakoid membrane is 3 pH units.
C) The luminal side of the thylakoid membrane has a higher pH than the stromal side.
D) The number of ATPs formed per oxygen molecule is about three.
E) The reaction centers, electron carriers, and ATP-forming enzymes are located in the thylakoid membrane.

ATP synthesis by photophosphorylation
Page: 787 Difficulty: 2 Ans: C

42. Cyclic electron flow in chloroplasts produces:

A) ATP and O_2, but not NADPH.
B) ATP, but not NADPH or O_2.
C) NADPH and ATP, but not O_2.
D) NADPH, but not ATP or O_2.
E) O_2, but not ATP or NADPH.

ATP synthesis by photophosphorylation
Page: 759 Difficulty: 3 Ans: B

43. Which of the following is *not* found in modern cyanobacteria?

A) PSI
B) PSII
C) NADH dehydrogenase complex I
D) Oxygen-reducing complex IV
E) Mitochondria

The evolution of oxygenic photosynthesis
Page: 759 Difficulty: 1 Ans: E

Short Answer Questions

44. As you read and answer this question, you are (presumably) consuming oxygen. What single reaction accounts for most of your oxygen consumption?

Ans: O_2 is converted to H_2O by electrons from the respiratory chain. The final step is the one catalyzed by cytochrome oxidase (complex IV).

Electron-transfer reactions in mitochondria
Page: 732 Difficulty: 1

45. Show the path of electrons from ubiquinone (Q or coenzyme Q) to oxygen in the mitochondrial respiratory chain. One of the two compounds (Q and O_2) has a standard reduction potential (E'°) of 0.82 V, and the other, 0.045 V. Which value belongs to each compound? How did you deduce this?

Ans: $QH_2 \rightarrow$ cyt $b \rightarrow$ cyt $c_1 \rightarrow$ cyt $c \rightarrow$ cyt $(a + a_3) \rightarrow O_2$

E'° for O_2 must be the larger positive value (+0.82) because electron flow occurs spontaneously to the electron acceptor with the more positive E'°.

Electron-transfer reactions in mitochondria
Page: 737 Difficulty: 3

46. Diagram the path of electron flow from NADH to the final electron acceptor during electron transport in mitochondria. For each electron carrier, indicate whether only electrons, or both electrons and protons, are accepted/donated by that carrier. Also, indicate where electrons from succinate oxidation enter the chain of carriers.

Ans: NADH (both) $\rightarrow$ FP (both) $\rightarrow$ Q (both) $\rightarrow$ cyt b (e^- only) $\rightarrow$ cyt c_1 (e^- only) $\rightarrow$ cyt c (e^- only) $\rightarrow$ cyt $(a + a_3)$ (e^- only) $\rightarrow O_2$ (both)

Electrons from succinate enter at Q.

Electron-transfer reactions in mitochondria
Pages: 737 Difficulty: 3

47. During electron transfer through the mitochondrial respiratory chain, the overall reaction is: $NADH + 1/2\ O_2 + H^+ \rightarrow NAD^+ + H_2O$. The difference in reduction potentials for the two half-reactions ($\Delta E'^{\circ}$) is +1.14 V. Show how you would calculate the standard free-energy change, $\Delta G'^{\circ}$, for the reaction as written above. (The Faraday constant, $\mathfrak{F}$, is 96.48 kJ/V·mol.)

Ans: $\Delta G'^{\circ} = -n\mathfrak{F}\Delta E'^{\circ} = (-2)(96.48\ \text{kJ/V·mol})(1.14\text{V}) = -220\ \text{kJ/mol}$

Electron-transfer reactions in mitochondria
Page: 737 Difficulty: 2

48. The standard reduction potential for ubiquinone (Q or coenzyme Q) is 0.045 V, and the standard reduction potential (E'°) for FAD is –0.219 V. Using these values, show that the oxidation of $FADH_2$ by ubiquinone theoretically liberates enough energy to drive the synthesis of ATP. The Faraday constant, $\mathfrak{F}$, is 96.48 kJ/V·mol. $\Delta G'^{\circ}$ for ATP synthesis is +30.5 kJ/mol.

Ans: $\Delta E'^{\circ} = E'^{\circ}(\text{oxidant}) - E'^{\circ}(\text{reductant}) = 0.045 - (-0.219) = +0.264$ V
$\Delta G'^{\circ} = -n\,\mathfrak{F}\Delta E'^{\circ} = (-2)(96.48\ \text{kJ/V·mol})(0.264\ \text{V}) = -51.0\ \text{kJ/mol}.$

Passage of two electrons from $FADH_2$ to Q can, in principle, provide sufficient energy for synthesizing 1 ATP.

Electron-transfer reactions in mitochondria
Page: 737 Difficulty: 2

49. A recently discovered bacterium carries out ATP synthesis coupled to the flow of electrons through a chain of carriers to some electron acceptor. The components of its electron transfer chain differ from those found in mitochondria; they are listed below with their standard reduction potentials.

Electron carriers in the newly discovered bacterium:

Oxidant	Reductant	Electrons transferred	E'° (V)
NAD^+	NADH	2	–0.32
flavoprotein *b* (FP_b) (oxidized)	flavoprotein *b* (reduced)	2	–0.62
cytochrome *c* (Fe^{3+})	cytochrome *c* (Fe^{2+})	1	+0.22
Fe-S protein (oxidized)	Fe-S protein (reduced)	2	+0.89
flavoprotein *a* (FP_a) (oxidized)	flavoprotein *a* (reduced)	2	+0.77

(a) Place the electron carriers in the order in which they are most likely to act in carrying electrons. (b) Is it likely that O_2 (for which $E'^\circ = 0.82$ V) is the final electron acceptor in this organism? Why or why not? (c) How would you calculate the maximum number of ATP molecules that could theoretically be synthesized, under standard conditions, per pair of electrons transfered through this chain of carriers? (The Faraday constant, $\mathfrak{F}$, is 96.48 kJ/V·mol.) $\Delta G'^\circ$ for ATP synthesis is +30.5 kJ/mol.

Ans: (a) $FP_b \rightarrow NAD^+ \rightarrow$ cyt *c* $\rightarrow FP_a \rightarrow$ Fe-S
(b) No; Fe-S has a larger E'°, so will probably be the terminal acceptor.
(c) First, calculate $\Delta G'^\circ$ for e^- flow from FP_b to Fe-S:
$\Delta E'^\circ = E'^\circ(\text{oxidant}) - E'^\circ(\text{reductant}) = +0.89 - (-0.62) = +1.51$ V
$\Delta G'^\circ = -n\mathfrak{F}\Delta E'^\circ = (-2)(96.48 \text{ kJ/V·mol})(1.51 \text{ V}) = -291$ kJ/2e^-

Theoretically, the flow of two electrons from FP_b to Fe-S could drive the synthesis of 291 kJ/30.5 kJ/mol = 9.5 mol ATP. Because only whole numbers of molecules can be made, the correct answer is 9 mol ATP per electron pair.

Electron-transfer reactions in mitochondria
Pages: 737 Difficulty: 3

50. Describe, in simple diagrams and a few words, the chemiosmotic theory for coupling oxidation to phosphorylation in mitochondria.

Ans: There are two central elements in the chemiosmotic model:
(1) Energetically favorable electron flow through asymmetrically arranged membrane-bound carriers causes transmembrane flow of H^+, creating a proton gradient (a proton motive force).
(2) The energy released by "downhill" movement of protons is captured when ADP and P_i are

condensed by ATP synthase (F_oF_1). (See Fig. 19-19.)

ATP synthesis
Pages: 748–749 **Difficulty: 2**

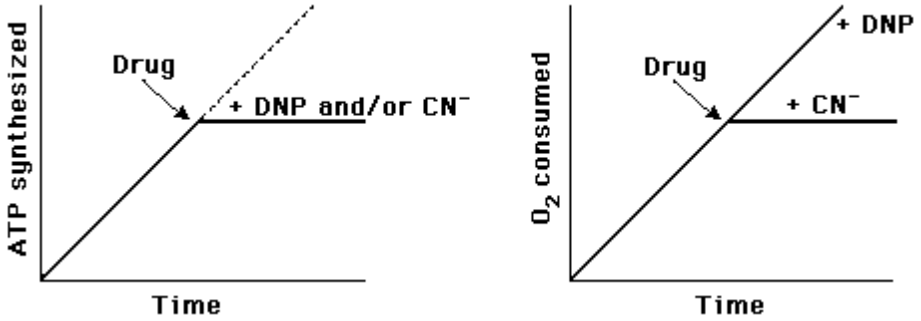
51. Compound X is an inhibitor of mitochondrial ATP synthesis. It was observed that when compound X was added to cells, the NAD^+/NADH ratio decreased. Would you expect X to be an uncoupling agent or an inhibitor of respiratory electron transfer? Explain.

    Ans: It is an inhibitor of electron transfer; its addition lowers the NAD^+/NADH ratio because NADH produced by oxidative reactions in mitochondria can no longer be reoxidized by electron flow to O_2. If it had been an uncoupling agent, the NADH could be reoxidized, the ratio of NAD^+/NADH would be unchanged, but there would be no build-up of a proton gradient (i.e., no synthesis of ATP).

ATP synthesis
Pages: 748–749 **Difficulty: 3**

52. Cyanide ion (CN^-) blocks electron transfer in mitochondria at the level of cytochrome $a + a_3$. 2,4-Dinitrophenol (DNP) is an uncoupler of mitochondrial oxidative phosphorylation. Draw graphs (ATP synthesized vs. time and O_2 consumed vs. time) with labeled curves to show the effects of adding each of these compounds *separately* to a suspension of mitochondria supplied with O_2, succinate, ADP, and P_i. Use an arrow to indicate the time of drug addition.

    Ans:



ATP synthesis
Pages: 748–749 **Difficulty: 3**

53. Give an example of (a) an uncoupler of oxidative phosphorylation, and (b) an inhibitor of respiration. (c) Describe the difference in the effects of such uncouplers and inhibitors on mitochondrial function.

    Ans: (a) Uncouplers include DNP, valinomycin, and CCCP. (b) Respiration inhibitors include antimycin A, piericidin A, CN^-, rotenone, and amytal. (c) Uncouplers stop formation of ATP while allowing electron transfer to continue. Inhibitors of respiration block both electron transfer and phosphorylation.

ATP synthesis
Pages: 748–749 **Difficulty: 2**

54. Mitochondria carrying out oxidative phosphorylation consume oxygen. Explain what happens to this oxygen, and describe the effect of an uncoupling agent such as 2,4-dinitrophenol on the rate of oxygen consumption. Assume there is a sufficient supply of oxidizable substrate, ADP, and P_i.

Ans: O_2 is reduced to H_2O by electrons passed through the respiratory chain. Addition of 2,4-dinitrophenol, which uncouples phosphorylation (ATP synthesis) from electron flow, actually stimulates O_2 consumption slightly by removing the "drag" of ATP synthesis.

ATP synthesis
Pages: 748–749 Difficulty: 2

55. The compound 2,4-dinitrophenol (DNP), an uncoupler, was briefly used as a weight-loss drug. Some of its effects in people who took the drug included weight loss and higher than normal body temperature. Some people even died. Explain the first two effects of the compound in biochemical terms.

Ans: The uncoupler allows continuous oxidation of fuels (like fatty acids from adipose tissue). Because little or no ATP synthesis occurs, the respiratory control that normally limits fuel oxidation when ATP levels are high does not operate. The energy of fuel oxidation is dissipated as heat, accounting for the elevated body temperature.

ATP synthesis
Pages: 748–749 Difficulty: 3

56. Although molecular oxygen (O_2) does not participate directly in any of the reactions of the citric acid cycle, the cycle operates only when O_2 is present. Explain this observation.

Ans: The citric acid cycle produces NADH, which is normally reoxidized to NAD^+ by the passage of electrons through the respiratory chain to O_2. With no O_2 to accept electrons, NADH accumulates, NAD^+ is depleted, and the citric acid cycle slows for lack of NAD^+.

ATP synthesis
Pages: 748–749 Difficulty: 3

57. The skunk cabbage (*Symphocarpus foetidus*) can maintain a temperature of 10–25 °C higher than the temperature of the surrounding air. Suggest a mechanism for this.

Ans: One possible heat-generating mechanism is partially uncoupled mitochondria in which some of the energy of electron flow is dissipated as heat. This is most easily achieved using bypasses of some of the mitochondrial complexes that normally capture favorable electron flow towards making a proton gradient that then drives ATP synthesis. If some of these complexes are omitted, then larger downhill steps by electrons that do not drive proton flow will instead generate heat. (See Box 19-1.)

ATP synthesis
Page: 747 Difficulty: 2

58. Consider a liver cell carrying out the oxidation of glucose under aerobic conditions. Suppose that we added a very potent and specific inhibitor of the mitochondrial ATP synthase, completely inhibiting this enzyme. Indicate whether each of the following statements about the effect of this inhibitor is true or false; if false, explain in a sentence or two why it is false.

____ (a) ATP production in the cell will quickly drop to zero.
____ (b) The rate of glucose consumption by this cell will decrease sharply.
____ (c) The rate of oxygen consumption will increase.
____ (d) The citric acid cycle will speed up to compensate.
____ (e) The cell will switch to fatty acid oxidation as an alternative to glucose oxidation, and the inhibitor will therefore have no effect on ATP production.

Ans: (a) False. Mitochondrial ATP synthesis will cease, but to compensate, cells will accelerate the production of ATP by glycolysis, preventing ATP levels from dropping to zero.
(b) False. The acceleration of glycolysis noted above will actually increase the rate of glucose consumption.
(c) False. Because electron transfer through the respiratory chain is tightly coupled to ATP synthesis, blocking ATP synthase blocks electron flow and oxygen consumption.
(d) False. The citric acid cycle is an *oxidative* pathway, producing NADH. When electron flow from NADH to O_2 is blocked, NADH accumulates, NAD^+ is depleted, and the citric acid cycle slows for lack of an electron acceptor (NAD^+).
(e) False. Oxidation of fats produces NADH, $FADH_2$, and acetyl-CoA, which is further oxidized via the citric acid cycle. For the reasons noted above, blocking electron flow through the respiratory chain prevents ATP synthesis with energy from fatty acid oxidation.

ATP synthesis
Pages: 748–749 **Difficulty: 3**

59. When the F_1 portion of the ATP synthetase complex is removed from the mitochondrial membrane and studied in solution, it functions as an ATPase. Why does it not function as an ATP synthetase?

Ans: Like all enzymes, the F_1 subunit of the ATP synthase catalyzes a reaction in *both* directions:

$$ADP + P_i \leftrightarrow ATP + H_2O$$

The standard free-energy change ($\Delta G'^{\circ}$) for ATP hydrolysis is –30.5 kJ/mol. With no proton motive force to drive the reaction toward ATP synthesis, the hydrolysis (ATPase activity) occurs spontaneously.

ATP synthesis
Pages: 750–752 **Difficulty: 2**

60. When the $\Delta G'^{\circ}$ of the ATP synthesis reaction is measured on the surface of the ATP synthase enzyme, it was found to be close to zero. Describe briefly why this is so.

Ans: The enzyme binds ATP more tightly than ADP thus stabilizing the former (i.e., the product of the synthesis reaction) relative to the latter (i.e., the reactant in the synthesis reaction).

ATP synthesis
Page: 751 **Difficulty: 2**

61. Explain briefly the current model for how the proton motive force that is generated by electron transport is used to drive the ATP synthesis reaction.

Ans: The enzyme binds ATP more tightly than ADP thus stabilizing the former, making the $\Delta G'^{\circ}$ of the

synthetic reaction more favorable. Once the reaction has occurred, the ATP product must be released from the enzyme (which is unfavorable). The proton motive force causes protons to move across the inner mitochondrial membrane through the pore in the F_o complex. This movement leads to conformational changes that decrease the affinity of the F_1 portion of the synthase for ATP, resulting in its release from the enzyme.

ATP synthesis
Pages: 750–755 Difficulty: 3

62. For each of the following explain whether they are transported into or out of the mitochondrion and comment briefly on the mechanism for each:

    (1) NADH
    (2) Inorganic phosphate
    (3) ADP

    Ans: NADH produced by glycolysis is indirectly transferred into the mitochondrion via the asparate-malate shuttle. (Only NADH equivalents are moved in, not the actual NADH molecule itself.) Inorganic phosphate is moved into the mitochondrion via the phosphate translocon, which accompanied by anti-port of a proton and is driven by the proton motive force. ADP is transported into the mitochondrion by the adenine nucleotide translocase that simultaneously moves an ATP out of the mitochondrion and is driven by the proton motive force.

ATP synthesis
Pages: 757–759 Difficulty: 2

63. Describe and explain the two different *ratios* that affect the rate of respiration in mitochondria. What is accomplished by these control mechanisms?

    Ans: The *acceptor control ratio* describes how much respiration takes place in the presence of ADP vs. the absence of ADP. (The availability of ADP to form ATP limits respiration.) The *mass-action ratio* relates the relative concentration of ATP to ADP. For both ratios, high ADP levels lead to increases in O_2 consumption. The overall effect is to keep the level of ATP nearly constant and to prevent oxidation of fuels when the demand for ATP is low.

Regulation of oxidative phosphorylation
Page: 760 Difficulty: 2

64. In his studies of alcoholic fermentation by yeast, Louis Pasteur noted that the sudden addition of oxygen (O_2) to a previously anaerobic culture of fermenting grape juice resulted in a dramatic decrease in the rate of glucose consumption. This "Pasteur effect" can be counteracted by the addition of 2,4-dinitrophenol (DNP), an uncoupler of oxidative phosphorylation. (a) Why would the yeast cells consume less glucose in the presence of oxygen? Can you estimate how much less glucose they would use? (b) Why would DNP counteract or prevent the Pasteur effect?

    Ans: (a) During aerobic oxidation of glucose, the yield of ATP is about 38 mol per mole of glucose. Under anaerobic conditions, only 2 ATP are produced per glucose. Thus, to get the same amount of ATP, a cell needs to use 19 times more glucose anaerobically. (b) Because DNP blocks the principal path of aerobic ATP production, aerobic cells in the presence of DNP need about the same amount of glucose as anaerobic cells.

Regulation of oxidative phosphorylation
Pages: 759 Difficulty: 2

65. Describe the role of hypoxia-inducible factor (HIF-1) in reducing reactive oxygen species (ROS).

    Ans: Under conditions of low oxygen, the electron transport chain may accidentally generate ROS because of the low levels of the final electron acceptor (oxygen). Therefore, HIF-1 is induced and serves as a transcription factor to increase glycolysis (i.e., anaerobic consumption of glucose). Some of the proteins whose transcription are turned on to effect this response are the glucose transporter, glycolytic enzymes, and lactate dehydrogenase. Additionally, the activity of pyruvate dehydrogenase is reduced (via phosphorylation) to slow the citric acid cycle and the flux of electrons into the electron transport chain.

Regulation of oxidative phosphorylation
Pages: 760–761 Difficulty: 3

66. Describe the role of cytochrome *c* in apoptosis.

    Ans: Upon receiving one or more of a variety of signals to trigger apoptosis, the permeability of the outer membrane of mitochondria increase, allowing the escape of cytochrome *c* from the intermembrane space. Cytochrome *c* then associates with Apaf-1 to form an apoptosome. The apopotsome provides the scaffold for the caspases that initiate a cascade of proteolytic activations, ultimately leading to cell death.

Mitochondria in thermogenesis, steroid synthesis, and apoptosis
Pages: 764–765 Difficulty: 2

67. Mutations in mitochondrial genes frequently produce diseases that affect the brain and skeletal muscle (mitochondrial encephalomyopathies). Why are these two tissues particularly sensitive to mitochondrial mutations?

    Ans: Both the brain and skeletal muscles are heavily dependent on an abundant supply of ATP for normal functioning.

Mitochondrial genes: their origin and the effects of mutation
Page: 767 Difficulty: 2

68. Discuss three lines of evidence that support the theory that mitochondria evolved from endosymbiontic bacteria.

    Ans: 1) Modern prokaryotes have respiratory chains very similar to those in mitochondria. 2) Bacteria translate protons outward across their plasma membrane and use this proton gradient to drive ATP synthesis and other energy-requiring processes such as flagellar motion. 3) Unlike most other organelles, mitochondria in eukaryotic cells retain their own genome.

Mitochondrial genes: their origin and the effects of mutation
Pages: 765–766 Difficulty: 2

69. Photophosphorylation differs from oxidative phosphorylation in that the former requires the input of energy in the form of _____ to create a good electron donor. In photophosphorylation, electrons

flow through a series of membrane-bound carriers including _____ , _____ , and _____ proteins, whereas _____ are pumped across a membrane to create an _____ potential.

Ans: Light; cytochromes; quinones; iron-sulfur; protons; electrochemical

General features of photophosphorylation
Pages: 769–770 Difficulty: 2

70. Describe the effect(s) that a mitochondrial uncoupler such as 2,4-dinitrophenol (DNP) would have on photophosphorylation.

Ans: Uncouplers like DNP, which act by dissipating transmembrane proton gradients, would uncouple photophosphorylation because it is also dependent on a transmembrane proton gradient to provide the energy required to synthesize ATP from ADP and P_i.

General features of photophosphorylation
Pages: 769–770Difficulty: 2

71. Discuss how "accessory pigments" are able to extend the range of light absorption of the chlorophylls. Name some accessory pigments.

Ans: Accessory pigments are able to absorb light and transfer the absorbed energy to the chlorophylls in a process know as "exciton transfer." Two accessory pigments are β-carotene and lutein.

Light absorption
Page: 773 Difficulty: 2

72. What is an action spectrum, and what do peaks in an action spectrum signify? Show a typical action spectrum plot for photosynthesis.

Ans: An action spectrum is a plot of the effectiveness of a process (such as photosynthesis) versus wavelength of incident light. Its peaks signify the presence of a chromophore that absorbs light at that wavelength. (See Fig. 19-53.)

Light absorption
Page: 774 Difficulty: 2

73. Describe what happens at photosystem I from the point where an antenna chlorophyll molecule absorbs a photon of light to the passage of an electron to $NADP^+$.

Ans: The antenna chlorophyll molecule passes the energy of the photon, via exciton transfer, to neighboring chlorophyll molecules and ultimately to reaction center chlorophyll molecules. This excites P700 to P700*, which donates an electron to A_0. From A_0, electrons pass to phylloquinone (A_1), through an Fe-S protein, to ferredoxin, then through a flavoprotein to $NADP^+$.

$$P700^* \rightarrow A_0 \rightarrow A_1 \rightarrow \text{Fe-S} \rightarrow NADP^+$$ (See Figs. 19-58; 19-60.)

The central photochemical event: light-driven electron flow

Pages: 779–781 **Difficulty: 2**

74. Give five general classes of electron carriers that function in both mitochondrial electron transfer to O_2 and photosynthetic electron transfer.

    Ans: Any five of the following: pyridine nucleotides (NADH, NADPH); flavin nucleotides ($FADH_2$, $FMNH_2$); quinones (ubiquinone, plastoquinone); cytochromes; Fe-S proteins; flavoproteins.

The central photochemical event: light-driven electron flow
Pages: 776–783 **Difficulty: 2**

75. The processes of oxidative phosphorylation coupled with electron transfer (in mitochondria) and photophosphorylation (in chloroplasts) resemble each other in certain respects. Describe five ways in which the two processes are similar, and describe three significant differences between the two processes.

    Ans: Similarities include: both contain a chain of membrane-bound electron carriers; electron transfer leads to establishment of a proton gradient; an ATPase/ATP synthase is a coupling factor; ATP synthesis is sensitive to uncouplers; both require a system of intact membranes to separate electrons inside and outside. Differences include: source of reducing power (NADH vs. light), product (ATP in respiration; NADPH in photosynthesis); source of oxidizing power (O_2 in mitochondria; light in photosynthesis).

The central photochemical event: light-driven electron flow
Pages: 769–783 **Difficulty: 3**

76. Show the path of electrons from photosystem II to NADPH in the chloroplast. What is the source of the energy that moves electrons through this path? Show where oxygen is involved in this pathway.

    Ans: $H_2O \rightarrow$ P680 $\rightarrow$ Pheophytin $\rightarrow$ PQA $\rightarrow$ PQB $\rightarrow$ cyt b_6f complex $\rightarrow$ Plastocyanin $\rightarrow$ P700 $\rightarrow A_0 \rightarrow A_1 \rightarrow$ Fe-S $\rightarrow$ Ferredoxin $\rightarrow$ Ferredoxin-$NADP^+$ oxidoreductase $\rightarrow NADP^+$.

    The energy that drives the electron flow is from light. O_2 is generated from H_2O when H_2O donates electrons to PSII.

The central photochemical event: light-driven electron flow
Pages: 776–779 **Difficulty: 2**

77. Plants carrying out photosynthesis produce O_2. Describe the source of this O_2, and explain, with chemical equations or schematic diagrams, why O_2 production occurs only during daylight hours.

    Ans: O_2 is generated by the splitting and oxidation of H_2O, driven by the absorption of a photon by PSII (which occurs only in daylight). (See Fig. 19-58.)

The central photochemical event: light-driven electron flow
Page: 779 **Difficulty: 2**

78. During photophosphorylation in plants, electrons flow through a series of carriers in the chloroplast. What is the ultimate donor of electrons, and what is the ultimate acceptor? What provides the energy to move those electrons?

Ans: The ultimate donor is H_2O, and the acceptor, $NADP^+$. The energy that drives this electron flow is from light.

The central photochemical event: light-driven electron flow
Page: 779 Difficulty: 2

79. Describe what happens when a photon is absorbed by photosystem II; end the description of electron flow at plastoquinone.

Ans: Absorption of a photon by PSII excites P680 to P680*, which passes an electron to pheophytin, which passes the electron to plastoquinone. P680, now lacking an electron, takes one away from a "water-splitting complex" of PSII, which in turn takes one from H_2O. (See Fig. 19-58.)

The central photochemical event: light-driven electron flow
Page: 779 Difficulty: 3

80. DCMU is an herbicide that acts by blocking photosynthetic electron flow from photosystem II (PSII) to the cytochrome b_6f complex. Predict the effect of DCMU on O_2 production and on ATP synthesis in the chloroplasts of plants sensitive to DCMU.

Ans: DCMU blocks electron flow from PSII to PSI, preventing generation of O_2 from H_2O. ATP synthesis will be inhibited, but not completely; cyclic photophosphorylation can continue in the presence of DCMU.

ATP synthesis by photophosphorylation
Pages: 739, 787Difficulty: 3

81. Given that ~8 photons are required to produce one molecule of O_2, roughly how many photons are required to produce one molecule of glucose.

Ans: Since the overall reaction of photosynthesis is $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$, six molecules of O_2 must be produced, requiring 6 x 8 or roughly 48 photons.

ATP synthesis by photophosphorylation
Page: 787 Difficulty: 2

82. Some photosynthetic bacteria contain the protein bacteriorhodopsin, which absorbs light and pumps protons out of the cell directly. Briefly describe how such a cell could use bacteriorhodopsin and an H^+ATPase to make ATP using light.

Ans: On exposure to light, bacteriorhodopsin would pump protons out of the cell, establishing a proton gradient. This proton gradient could be used by an H^+ATPase to generate ATP.

The Evolution of Oxygenic Photosynthesis
Pages: 789–790 Difficulty: 1